Higher partial derivatives

$$(f_x)_x = f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) \qquad (f_x)_y = f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$

$$(f_y)_x = f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) \qquad (f_y)_y = f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$$

} second derivatives

In general

$$u = f(x_1, x_1, \cdots, x_n)$$

$$\rightarrow \frac{\partial f}{\partial x_i} = f_{x_i} = D_i f = \frac{\partial u}{\partial x_i}, \quad i = 1, 2, \cdots, n.$$

Clairaut's Theorem

If f_{xy} and f_{yx} are both continuous at (a, b), then $f_{xy}(a,b) = f_{yx}(a,b)$.

§ 14.5

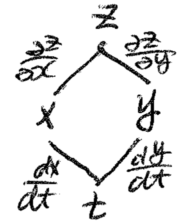
Chain Rules

Case I:

If $z = f(x, y)$ with $x = g(t)$ and $y = h(t)$, then

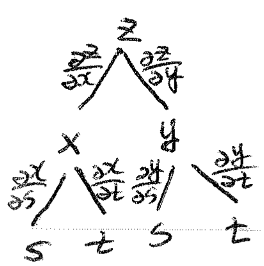
$$\boxed{\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} \\ &= f_x g' + f_y h'\end{aligned}}$$

Tree diagram



Case II:

If $z = f(x, y)$ with $x = g(s, t)$ and $y = h(s, t)$, then



$$\boxed{\begin{aligned}\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s} & \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} \\ &= f_x g_s + f_y h_s & &= f_x g_t + f_y h_t\end{aligned}}$$

General form

$u(x_1, x_2, \cdots, x_n)$, $x_j(t_1, t_2, \cdots, t_m)$ for $j = 1, 2, \cdots, n$

then

$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1}\frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2}\frac{\partial x_2}{\partial t_i} + \cdots + \frac{\partial u}{\partial x_n}\frac{\partial x_n}{\partial t_i}$$

$i = 1, 2, \cdots, n$.

Implicit differentiation

$$F(x,y)=0 \longrightarrow y=f(x) \text{ is implicitly defined}$$

$$\frac{\partial F}{\partial x}+\frac{\partial F}{\partial y}\frac{\partial y}{\partial x}=0$$

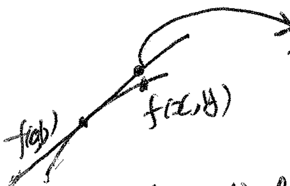
$$\frac{\partial y}{\partial x}=-\frac{\partial F/\partial x}{\partial F/\partial y}=-\frac{F_x}{F_y}$$

§14.4.

Given a surface S $(z=f(x,y))$, its tangent plane at the point $(x_0, y_0, \underset{\overset{\|}{f(x_0,y_0)}}{z_0})$ is defined by

$$z-z_0=f_x(x_0,y_0)(x-x_0)+f_y(x_0,y_0)(y-y_0)$$

Linear Approximation



$$f(x,y)\approx \underline{f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-b)}=:L(x,y).$$

differentials $\quad z=f(x,y)$

$$\underline{dz=f_x(x,y)dx+f_y(x,y)dy}.$$

total differentials. $\rightarrow f(x+dx,y+dy)\approx f(x,y)+dz$

§14.6.

The directional derivative of f at (x,y) in the direction of a unit vector $u=\langle u_1,u_2\rangle$ is

$$\underline{D_u f(x,y)}=\lim_{h\to 0}\frac{f(x+hu_1,y+hu_2)-f(x,y)}{h}$$

$$=\underbrace{\langle f_x(x,y),\ f_y(x,y)\rangle}_{\nabla f(x,y)\ \text{— Gradient vector}}\cdot\langle u_1,u_2\rangle$$

$$=\underline{\nabla f\cdot u}=f_xu_1+f_yu_2$$

Function of three variables: $D_u f(x,y,z)=\langle f_x,f_y,f_z\rangle\cdot\langle u_1,u_2,u_3\rangle$

$$=\nabla f\cdot u=f_xu_1+f_yu_2+f_zu_3$$

$\max_u D_u f = |\nabla f| \to$ when take u be the same direction as ∇f.

Tangent planes to level surfaces

Level surface: $F(x,y,z)=k$

At (x_0,y_0,z_0), normal vector is given by $\nabla f(x_0,y_0,z_0)$, thus the equation of the tangent plane to the level surface is given by.

$$F_x(x_0,y_0,z_0)(x-x_0)+F_y(x_0,y_0,z_0)(y-y_0)+F_z(x_0,y_0,z_0)(z-z_0)=0.$$

Normal line - symmetric equations

$$\frac{x-x_0}{F_x(x_0,y_0,z_0)} = \frac{y-y_0}{F_y(x_0,y_0,z_0)} = \frac{z-z_0}{F_z(x_0,y_0,z_0)}$$

§14.7.

local maximum/minimum. absolute maximum/minimum.

Critical points (or stationary points): $f_x(a,b)=0$, $f_y(a,b)=0$
$\Rightarrow (a,b)$ is a critical point.

First derivative Test:

If f has a local maximum/minimum at (a,b), then $f_x(a,b)=f_y(a,b)=0$, i.e., (a,b) is a critical point.

Second derivative Test

Suppose $f_x(a,b)=f_y(a,b)=0$. Let $D=D(a,b)=f_{xx}(a,b)f_{yy}(a,b)-[f_{xy}(a,b)]^2$.

(a) If $D>0$ and $f_{xx}(a,b)>0$, then $f(a,b)$ is a local minimum.

(b) If $D>0$ and $f_{xx}(a,b)<0$, then $f(a,b)$ is a local maximum.

(c) If $D<0$, then $f(a,b)$ is neither a local maximum nor minimum, (a,b) is called a saddle point.

§14.8.

$$\begin{cases} \max/\min \ f(x,y,z) \\ \text{s.t.} \quad \underbrace{g(x,y,z)=k}_{\text{constraint}} \end{cases}$$

Method of Lagrange multipliers

(a) Find all values of x, y, z and $\underset{\text{multiplier}}{\underline{\lambda}}$ such that

$$\begin{cases} \nabla f(x,y,z) = \lambda \nabla g(x,y,z) \\ g(x,y,z) = k. \end{cases} \quad \text{i.e.} \quad \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \\ f_z = \lambda g_z \\ g = k \end{cases}$$

(b). Evaluate f at all points resulted from (a). The largest is the maximum value of f and the smallest is the minimum value of f.

For Two constraints problem

$$\begin{cases} \max/\min \ f(x,y,z) \\ \text{s.t.} \ \ g(x,y,z) = k_1 \\ \qquad h(x,y,z) = k_2 \end{cases}$$

$$\Rightarrow \begin{cases} \nabla f = \underline{\lambda} \nabla g + \underline{\mu} \nabla h \\ g = k_1 \\ h = k_2 \end{cases}$$

λ, μ two multipliers.

Chapter 15.

§15.1.

The double integral of f over the rectangle R is

$$\iint_R f(x,y)\,dA = \underbrace{\lim_{m,n\to+\infty} \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_i^*, y_j^*)\,\underbrace{\Delta A}_{=\Delta x \Delta y}}_{\text{double Riemann sum}}$$

If $f(x,y) \geq 0$, the volume V of the solid that lies above R and below the surface $z = f(x,y)$ is

$$V = \iint_R f(x,y)\,dA.$$

Midpoint Rule:

$$\iint_R f(x,y)\,dA \approx \sum_{i=1}^{m}\sum_{j=1}^{n} f(\bar{x}_i, \bar{y}_j)\,\Delta A.$$

<u>Properties</u>

$$\iint_R [f+g]\,dA = \iint_R f\,dA + \iint_R g\,dA$$

$$\iint_R \alpha f\,dA = \alpha \iint_R f\,dA \quad \text{where } \alpha \text{ is a constant.}$$

§15.2.

<u>Fubini's Theorem</u>

$R = \{(x,y) \mid a \leq x \leq b,\ c \leq y \leq d\}$, f is continuous over R, then

$$\iint_R f(x,y)\,dA = \int_a^b \int_c^d f(x,y)\,dy\,dx = \int_c^d \int_a^b f(x,y)\,dx\,dy.$$

called "iterated integral"

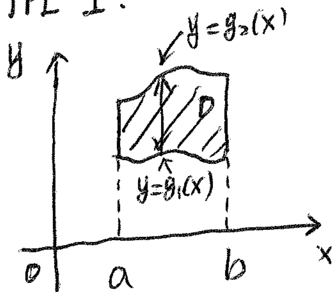
<u>special case</u> $f(x,y) = g(x)\,h(y)$ on $R = [a,b]\times[c,d]$

$$\iint_R f(x,y)\,dA = \int_a^b g(x)\,dx \int_c^d h(y)\,dy.$$

§15.3.

TYPE of Planar regions

TYPE I:

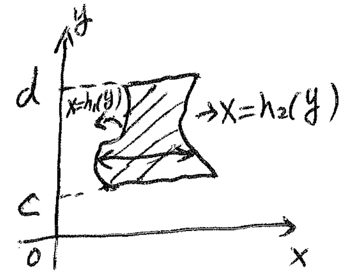


$D=\{(x,y) \mid a\le x\le b,\ g_1(x)\le y\le g_2(x)\}$.

$\Downarrow$

$$\iint_D f(x,y)\,dA=\int_a^b\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$

TYPE II:



$D=\{(x,y) \mid c\le y\le d,\ h_1(y)\le x\le h_2(y)\}$.

$\Downarrow$

$$\iint_D f(x,y)\,dA=\int_c^d\int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy.$$

Remark: The plane region D could be both Type I and Type II sometimes.
Always use the one that gives simpler integration formula.

Properties

(i) If $D=D_1\cup D_2$ and $\mathring{D}_1\cap\mathring{D}_2=\phi$, then

$$\iint_D f\,dA=\iint_{D_1} f\,dA+\iint_{D_2} f\,dA.$$

(ii) If $f\ge g$ on D, then

$$\iint_D f\,dA\ge\iint_D g\,dA.$$

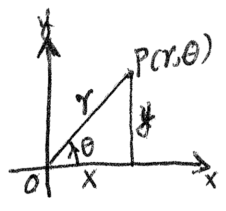
(iii) $\iint_D 1\,dA=\text{Area}(D)$. where D is a plane region.

(iv). If $m\le f(x,y)\le M$ on D, then

$$\underline{m\,\text{Area}(D)\le\iint_D f(x,y)\,dA\le M\,\text{Area}(D)}$$

↳ used for a rough estimation of the double integral $\iint_D f(x,y)\,dA$.

§15.4.



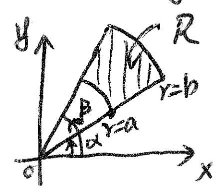
Polar coordinates (r, θ)

$$r^2 = x^2 + y^2, \quad \tan\theta = \frac{y}{x}$$

$$\Updownarrow$$

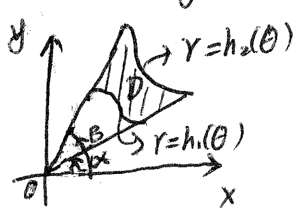
$$x = r\cos\theta, \quad y = r\sin\theta$$

Polar rectangle: $R = \{(r,\theta) \mid a \le r \le b,\ \alpha \le \theta \le \beta\}$.



$$\iint_R f(x,y)\,dA = \int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\,\underline{r\,dr\,d\theta}$$

General region: $D = \{(r,\theta) \mid \alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta)\}$.



$$\iint_D f(x,y)\,dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\,\underline{r\,dr\,d\theta}$$

<u>The area of D is then</u>

$$A(D) = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} r\,dr\,d\theta = \int_\alpha^\beta \frac{1}{2}\left(h_2^2(\theta) - h_1^2(\theta)\right) d\theta$$

§15.6.

<u>A rectangular box</u>: $B = \{(x,y,z) \mid a \le x \le b,\ c \le y \le d,\ r \le z \le s\}$.

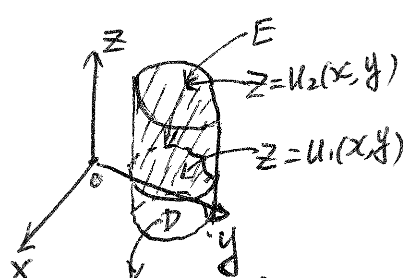
$$\iiint_B f(x,y,z)\,dV = \int_r^s \int_c^d \int_a^b f(x,y,z)\,dx\,dy\,dz$$

$$= \int_a^b \int_c^d \int_r^s f(x,y,z)\,dz\,dy\,dx$$

(change of intergration order for x, y, z)

$$= \cdots$$

General bounded E in 3D space



projection of E onto the xy plane.

<u>TYPE 1</u>: $E = \{(x,y,z) \mid (x,y) \in D,\ u_1(x,y) \le z \le u_2(x,y)\}$.

then $$\iiint_E f(x,y,z)\,dV = \iint_D \left[\int_{u_1(x,y)}^{u_2(x,y)} f(x,y,z)\,dz\right] dA$$

↳ reduce to a double integral

Now consider D – plane region

I) $D = \{(x,y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\} \Rightarrow$

$$\iiint_E f(x,y,z)\,dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y)}^{u_2(x,y)} f(x,y,z)\,dz\,dy\,dx$$

II) $D=\{(x,y)\mid c\le y\le d,\ h_1(y)\le x\le h_2(y)\}\Rightarrow$

$$\iiint_E f(x,y,z)\,dV=\int_c^d\int_{h_1(y)}^{h_2(y)}\int_{u_1(x,y)}^{u_2(x,y)} f(x,y,z)\,dz\,dx\,dy.$$

<u>TYPE 2</u>: $E=\{(x,y,z)\mid (y,z)\in D,\ u_1(y,z)\le x\le u_2(y,z)\}$, then

$$\iiint_E f(x,y,z)\,dV=\iint_D\Big[\int_{u_1(y,z)}^{u_2(y,z)} f(x,y,z)\,dx\Big]dA.$$

<u>TYPE 3</u>: $E=\{(x,y,z)\mid (x,z)\in D,\ u_1(x,z)\le y\le u_2(x,z)\}$, then

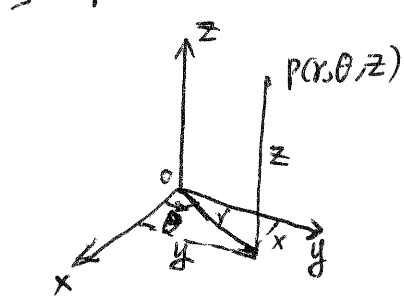
$$\iiint_E f(x,y,z)\,dV=\iint_D\Big[\int_{u_1(y,z)}^{u_2(y,z)} f(x,y,z)\,dy\Big]dA.$$

<u>Volume of E</u>: $V(E)=\iiint_E 1\,dV=\iiint_E dV$

Let $\rho(x,y,z)$ be the density function of the object occupying E, then its mass is

$$m=\iiint_E \rho(x,y,z)\,dV$$

§15.7.



Cylindrical coordinates $(r,\theta,z)\Leftrightarrow\{$Polar coordinates$\}+\{z\}$

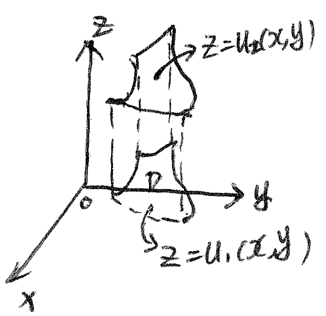
$$r^2=x^2+y^2,\quad \tan\theta=\frac{y}{x},\quad z=z$$

$$\Updownarrow$$

$$x=r\cos\theta,\quad y=r\sin\theta,\quad z=z.$$

Suppose E is a Type 1 region in the 3D space (§15.6)

$$E=\{(x,y,z)\mid (x,y)\in D,\ u_1(x,y)\le z\le u_2(x,y)\}$$



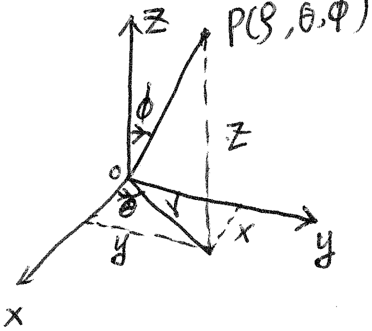
with

$D=\{(r,\theta)\mid \alpha\le\theta\le\beta,\ h_1(\theta)\le r\le h_2(\theta)\}\rightarrow$ polar coordinates

then

$$\iiint_E f(x,y,z)\,dV=\int_\alpha^\beta\int_{h_1(\theta)}^{h_2(\theta)}\int_{u_1(r\cos\theta,r\sin\theta)}^{u_2(r\cos\theta,r\sin\theta)} f(r\cos\theta,r\sin\theta,z)\,\underline{r\,dz\,dr\,d\theta}$$

§15.8.



spherical coordinate (ρ, θ, ϕ)

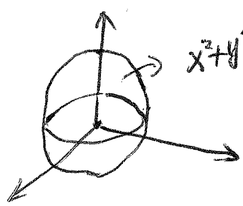
$\rho \geq 0, \quad 0 \leq \phi \leq \pi$.

$\rho = \sqrt{x^2+y^2+z^2}, \quad \tan\theta = \frac{y}{x}, \quad \cos\phi = \frac{z}{\sqrt{x^2+y^2+z^2}}$

$\Updownarrow$

$x = \rho\sin\phi\cos\theta, \ y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi$

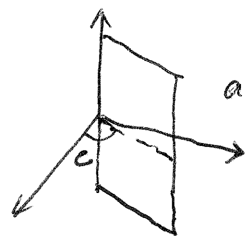
i) $\rho = C \quad (C > 0)$



$x^2+y^2+z^2 = C^2$

a sphere

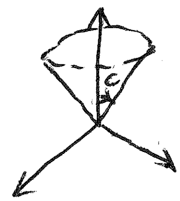
ii) $\theta = C$



a plane

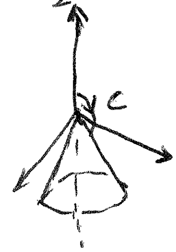
iii) $\phi = C$

$0 < C < \pi/2$

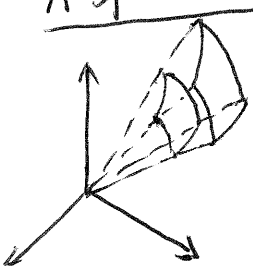


a half cone

$\frac{\pi}{2} < C < \pi$



A spherical wedge $B = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$, then



$$\iiint_B f(x,y,z)\,dV = \int_c^d \int_\alpha^\beta \int_a^b f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\,\underline{\rho^2\sin\phi\,d\rho\,d\theta\,d\phi}$$

General spherical region:

$E = \{(\rho, \theta, \phi) \mid \alpha \leq \theta \leq \beta, \ c \leq \phi \leq d, \ g_1(\theta,\phi) \leq \rho \leq g_2(\theta,\phi)\}$.

$$\Rightarrow \iiint_E f(x,y,z)\,dV = \int_c^d \int_\alpha^\beta \int_{g_1(\theta,\phi)}^{g_2(\theta,\phi)} f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\,\underline{\rho^2\sin\phi\,d\rho\,d\theta\,d\phi}$$